PUMP AI COPILOT RESPONSE HANDBOOK

Document ID: PUMP-AI-001

Revision: 1.0

Asset Class: Centrifugal Pump

Purpose: AI Reasoning Framework, Diagnostic Logic, Recommendation Engine and Response Standardization

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1. Purpose

This handbook defines how the Industrial AI Copilot shall:

- Interpret telemetry
- Interpret ML predictions
- Determine equipment health
- Perform root cause analysis
- Generate recommendations
- Explain reasoning
- Escalate critical conditions

The objective is to ensure AI responses remain consistent with refinery engineering practices.

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2. AI Decision Framework

The AI Copilot shall never make decisions using a single parameter.

Every conclusion shall consider:

Telemetry

+

Failure Probability

+

Failure Mode

+

Remaining Useful Life

+

Operational Context

+

Historical Cases

+

Maintenance Knowledge

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3. Inputs Available to AI

Telemetry

The AI receives:

bearing_temp

motor_temp

vibration

flow_rate

suction_pressure

discharge_pressure

motor_current

lubrication_level

seal_leakage_rate

Machine Learning Outputs

Failure Probability

Failure Mode

RUL

RAG Knowledge

Operations Manual

Maintenance Manual

FMEA

Troubleshooting Guide

Alarm Matrix

Bearing SOP

Cavitation Guide

RCA Reports

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4. AI Response Structure

Every diagnostic response should contain:

Section 1

Current Condition

Section 2

Evidence

Section 3

Probable Failure Mode

Section 4

Risk Assessment

Section 5

Recommended Actions

Section 6

Urgency

Section 7

Supporting References

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5. Pump Health Classification

HEALTHY

Conditions

Failure Probability:

<30%

RUL:

180 Days

No critical telemetry deviations.

AI Response

Pump is operating normally.

No significant degradation detected.

Continue routine monitoring.

DEGRADING

Conditions

Failure Probability:

30–60%

RUL:

90–180 Days

Minor telemetry deviations present.

AI Response

Early signs of degradation detected.

Maintenance planning is recommended.

Continued monitoring is advised.

AT RISK

Conditions

Failure Probability:

60–80%

RUL:

30–90 Days

Multiple telemetry anomalies detected.

AI Response

The pump is showing signs of progressive degradation.

Maintenance scheduling should be initiated.

CRITICAL

Conditions

Failure Probability:

80–95%

RUL:

10–30 Days

Significant telemetry abnormalities.

AI Response

Critical degradation detected.

Immediate reliability review required.

EMERGENCY

Conditions

Failure Probability:

95%

RUL:

<10 Days

Telemetry approaching trip conditions.

AI Response

Potential imminent failure.

Immediate maintenance intervention required.

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6. Bearing Failure Reasoning Rules

If:

Bearing Temperature ↑

AND

Vibration ↑

AND

Motor Current ↑

Then:

Diagnosis:

Bearing Failure

Confidence:

Very High

Additional Evidence

Lubrication ↓

Confidence increases further.

Expected Consequences

- Bearing seizure
- Shaft damage
- Seal damage

Recommended Actions

Inspect bearing.

Inspect lubrication.

Schedule replacement.

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7. Seal Leakage Reasoning Rules

If:

Seal Leakage ↑

AND

Flow Rate ↓

AND

Discharge Pressure ↓

Then:

Diagnosis:

Seal Leakage

Confidence:

Very High

Expected Consequences

- Product loss
- Safety risk
- Environmental risk

Recommended Actions

Inspect seal assembly.

Replace seal.

Verify shaft condition.

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8. Cavitation Reasoning Rules

If:

Suction Pressure ↓

AND

Flow Rate ↓

AND

Vibration ↑

Then:

Diagnosis:

Cavitation

Confidence:

Very High

Expected Consequences

- Impeller erosion
- Reduced efficiency
- Secondary bearing damage

Recommended Actions

Inspect suction system.

Inspect strainers.

Verify NPSH.

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9. RUL Interpretation Rules

RUL > 180 Days

Risk:

Low

Action:

Monitor

RUL 90–180 Days

Risk:

Moderate

Action:

Maintenance planning

RUL 30–90 Days

Risk:

High

Action:

Schedule maintenance

RUL 10–30 Days

Risk:

Critical

Action:

Reliability review

RUL <10 Days

Risk:

Emergency

Action:

Immediate intervention

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10. Failure Probability Interpretation Rules

Probability <30%

Normal

Probability 30–60%

Early degradation

Probability 60–80%

Developing failure

Probability 80–95%

High confidence failure progression

Probability >95%

Potential imminent failure

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11. AI Severity Scoring Model

Severity Score Components

Failure Probability:

40%

Telemetry Deviation:

30%

RUL:

20%

Asset Criticality:

10%

Severity Levels

1

Healthy

2

Degrading

3

At Risk

4

Critical

5

Emergency

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12. AI Escalation Logic

Generate Maintenance Notification

If:

Failure Probability >60%

Generate Reliability Alert

If:

Failure Probability >80%

AND

RUL <30 Days

Generate Critical Alert

If:

Failure Probability >90%

AND

RUL <15 Days

Generate Emergency Alert

If:

Failure Probability >95%

AND

RUL <10 Days

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13. AI Explanation Templates

Template A

Bearing Failure

Response:

The telemetry pattern indicates bearing degradation.

Evidence:

- Bearing Temperature increasing
- Vibration increasing
- Motor Current increasing

This pattern is consistent with historical bearing failure cases documented in refinery reliability records.

Recommended Action:

Inspect and replace bearing assembly.

Template B

Seal Leakage

Response:

The telemetry pattern indicates seal degradation.

Evidence:

- Leakage increasing
- Flow decreasing
- Discharge pressure decreasing

This condition may result in product loss and environmental exposure.

Recommended Action:

Inspect and replace seal assembly.

Template C

Cavitation

Response:

The telemetry pattern indicates cavitation.

Evidence:

- Low suction pressure
- Reduced flow
- Elevated vibration

This condition may cause impeller erosion and secondary bearing damage.

Recommended Action:

Inspect suction system and verify NPSH margin.

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14. AI Response Example

Input

Bearing Temperature:

91°C

Vibration:

8.7 mm/s

Motor Current:

143 A

Failure Probability:

94%

RUL:

16 Days

Expected AI Output

Current Condition:

Critical

Probable Failure Mode:

Bearing Failure

Evidence:

Bearing Temperature elevated.

Vibration elevated.

Motor Current elevated.

Risk Assessment:

High probability of continued degradation.

Potential bearing seizure if not addressed.

Recommended Actions:

Inspect bearing assembly.

Verify lubrication condition.

Schedule replacement.

Urgency:

Within 24–48 Hours

References:

Pump FMEA

Pump Maintenance Manual

RCA Bearing Failure Report

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15. Continuous Learning Rules

Whenever a confirmed failure occurs:

Capture:

Telemetry

Failure Mode

Root Cause

Corrective Action

Downtime

Lessons Learned

Store as:

RCA Knowledge Document

Add to:

RAG Knowledge Base

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16. Final Principle

The AI Copilot is a decision-support system.

It shall:

- Explain conclusions

- Cite evidence
- Reference engineering knowledge
- Recommend actions

It shall not replace engineering judgment.

Final maintenance decisions remain the responsibility of qualified personnel.

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17. Summary

The AI Copilot shall combine:

Telemetry

+

ML Predictions

+

RAG Knowledge

+

Historical Incidents

to produce transparent, explainable and actionable maintenance recommendations.

The goal is to detect failures early, reduce downtime, improve reliability and support refinery operations through intelligent engineering assistance.